Digital Design and Computer Organization Laboratory UE24CS251A

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Mini Project Report

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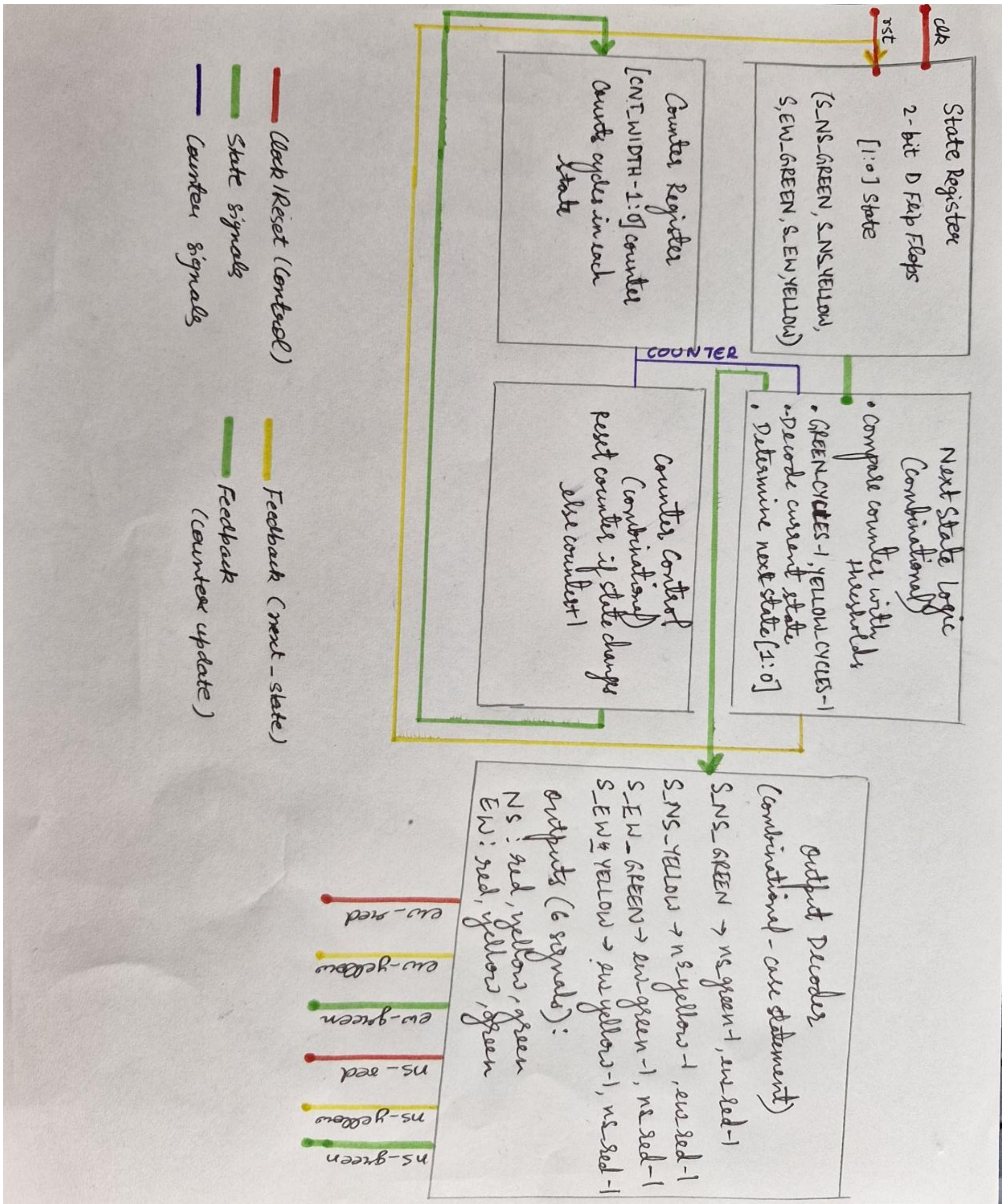
1. Problem Statement
   Implement the control logic for the traffic lights, handle timing constraints, and simulate the system's behavior.

2. Circuit Diagram/Block Diagram



3. Verilog Code Screenshot

- Design File:

```verilog
module traffic_controller #(
    parameter GREEN_CYCLES = 200,
    parameter YELLOW_CYCLES = 50
)(
    input wire clk,
    input wire rst,
    output reg ns_red, ns_yellow, ns_green,
    output reg ew_red, ew_yellow, ew_green
);
    localparam S_NS_GREEN  = 2'b00;
    localparam S_NS_YELLOW = 2'b01;
    localparam S_EW_GREEN  = 2'b10;
    localparam S_EW_YELLOW = 2'b11;
    reg [1:0] state = S_NS_GREEN;
    reg [1:0] next_state;
    reg [31:0] counter = 0;
    always @(*) begin
        case(state)
            S_NS_GREEN:
                next_state = (counter >= GREEN_CYCLES) ? S_NS_YELLOW : S_NS_GREEN;
            S_NS_YELLOW:
                next_state = (counter >= YELLOW_CYCLES) ? S_EW_GREEN : S_NS_YELLOW;
            S_EW_GREEN:
                next_state = (counter >= GREEN_CYCLES) ? S_EW_YELLOW : S_EW_GREEN;
            S_EW_YELLOW:
                next_state = (counter >= YELLOW_CYCLES) ? S_NS_GREEN : S_EW_YELLOW;
            default:
                next_state = S_NS_GREEN;
        endcase
    end
    always @(posedge clk) begin
        if (rst) begin
            state <= S_NS_GREEN;
            counter <= 0;
        end else begin
            if (state != next_state) begin
                state <= next_state;
                counter <= 0;
            end else begin
                counter <= counter + 1;
            end
        end
    end
    always @(*) begin
        ns_red = 0; ns_yellow = 0; ns_green = 0;
        ew_red = 0; ew_yellow = 0; ew_green = 0;
        case(state)
            S_NS_GREEN:  begin ns_green = 1; ew_red = 1; end
            S_NS_YELLOW: begin ns_yellow = 1; ew_red = 1; end
            S_EW_GREEN:  begin ew_green = 1; ns_red = 1; end
            S_EW_YELLOW: begin ew_yellow = 1; ns_red = 1; end
        endcase
    end
endmodule
```

- Test_bench File:

```
`timescale 1ns/1ps
module tb_traffic_controller;

    localparam CLK_PERIOD = 10;

    reg clk;
    reg rst;
    wire ns_red, ns_yellow, ns_green;
    wire ew_red, ew_yellow, ew_green;

    traffic_controller #(
        .GREEN_CYCLES(20),
        .YELLOW_CYCLES(5)
    ) DUT (
        .clk(clk),
        .rst(rst),
        .ns_red(ns_red),
        .ns_yellow(ns_yellow),
        .ns_green(ns_green),
        .ew_red(ew_red),
        .ew_yellow(ew_yellow),
        .ew_green(ew_green)
    );

    initial begin
        clk = 1'b0;
        forever #(CLK_PERIOD/2) clk = ~clk;
    end

    initial begin
        $dumpfile("traffic_tb.vcd");
        $dumpvars(0, tb_traffic_controller);

        rst = 1'b1;
        #(CLK_PERIOD*5);
        rst = 1'b0;

        #(CLK_PERIOD * 5000);

        $display("Simulation finished");
        $finish;
    end

    initial begin
        $display("Time(ns)  ns_G ns_Y ns_R  ew_G ew_Y ew_R");
        forever begin
            @(posedge clk);
            $display("%8t  %b     %b    %b     %b    %b    %b", $time,
                ns_green, ns_yellow, ns_red,
                ew_green, ew_yellow, ew_red);
        end
    end

endmodule
```

4. VVP Output Screen Shot

```
VCD info: dumpfile traffic_tb.vcd opened for output.
Time(ns)  ns_G ns_Y ns_R  ew_G ew_Y ew_R
    5000  1     0    0     0    0    1
   15000  1     0    0     0    0    1
   25000  1     0    0     0    0    1
   35000  1     0    0     0    0    1
   45000  1     0    0     0    0    1
   55000  1     0    0     0    0    1
   65000  1     0    0     0    0    1
   75000  1     0    0     0    0    1
   85000  1     0    0     0    0    1
   95000  1     0    0     0    0    1
  105000  1     0    0     0    0    1
  115000  1     0    0     0    0    1
  125000  1     0    0     0    0    1
  135000  1     0    0     0    0    1
  145000  1     0    0     0    0    1
  155000  1     0    0     0    0    1
  165000  1     0    0     0    0    1
  175000  1     0    0     0    0    1
  185000  1     0    0     0    0    1
  195000  1     0    0     0    0    1
  205000  1     0    0     0    0    1
  215000  1     0    0     0    0    1
  225000  1     0    0     0    0    1
  235000  1     0    0     0    0    1
  245000  1     0    0     0    0    1
  255000  1     0    0     0    0    1
  265000  0     1    0     0    0    1
  275000  0     1    0     0    0    1
  285000  0     1    0     0    0    1
  295000  0     1    0     0    0    1
  305000  0     1    0     0    0    1
  315000  0     1    0     0    0    1
  325000  0     0    1     1    0    0
  335000  0     0    1     1    0    0
  345000  0     0    1     1    0    0
  355000  0     0    1     1    0    0
  365000  0     0    1     1    0    0
  375000  0     0    1     1    0    0
```

```
49785000  1     0    0     0    0    1
49795000  1     0    0     0    0    1
49805000  1     0    0     0    0    1
49815000  1     0    0     0    0    1
49825000  1     0    0     0    0    1
49835000  1     0    0     0    0    1
49845000  1     0    0     0    0    1
49855000  1     0    0     0    0    1
49865000  1     0    0     0    0    1
49875000  1     0    0     0    0    1
49885000  1     0    0     0    0    1
49895000  1     0    0     0    0    1
49905000  1     0    0     0    0    1
49915000  1     0    0     0    0    1
49925000  1     0    0     0    0    1
49935000  1     0    0     0    0    1
49945000  0     1    0     0    0    1
49955000  0     1    0     0    0    1
49965000  0     1    0     0    0    1
49975000  0     1    0     0    0    1
49985000  0     1    0     0    0    1
49995000  0     1    0     0    0    1
50005000  0     0    1     1    0    0
50015000  0     0    1     1    0    0
50025000  0     0    1     1    0    0
50035000  0     0    1     1    0    0
50045000  0     0    1     1    0    0
Simulation finished
tb_traffic_controller.v:41: $finish called at 50050000 (1ps)
```

5. GTKWAVE Screenshot